Build Overview:

MECHANICAL MANUFACTURING INSTRUCTIONS

CDU Water Return Assembly

CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by
	1	New Issue	JOSEPH AMAD

Prepared By:
JOSEPH R. AMAD
Manufacturing Engineer

Checked By:
IVAN NOWELL PANIQUE
Team Lead

Approved By:
SHERWIN ARIOSTE
Operations Manager

SPECIAL TOOLS

-3/8”, 1-1/2”, 50mm Wrench
50mm Crowsfoot
14mm Hex wrench with 5.6 Nm Torque wrench
156 Nm Torque Wrench

Water Return Orientation Fixture (654-641-50)
Sense Line Bend Fixture (654-641-99)
1.6 Nm Torque Driver
4mm Hex Bit
Loctite 565 (TPN: 636-666-00)

METHOD:

Flow chart of operation steps

SIC preparation
↓
Preparation Tools
↓
Prepare Materials
↓
Complete Operation
↓
Inventory Materials
↓
Operation Check
↓
Inventory tool

Size Hardware	Newton Meter
M2 x 0.4	0.2
M3 x 0.5	0.7
M4 x 0.7	1.6
M5 x 0.8	3.3
M6 x 1.0	5.6
M8 x 1.25	13.5

Torque Rules

- Torque values shown are for steel-to-steel material only.
- Torque all screws and nuts unless otherwise specified.
- Torque table is for reference. See MFE-500-01 for detailed information.

MAN: CERTIFIED OPERATOR, TECHNICIAN

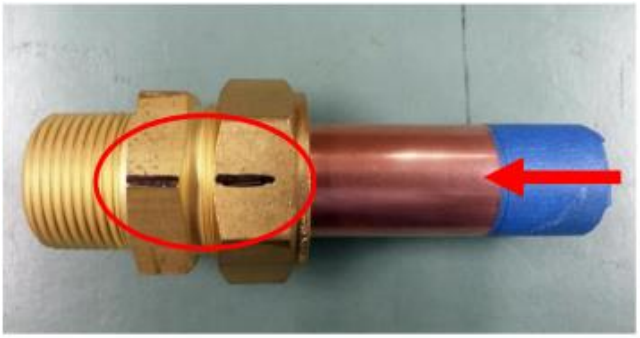
NOTES

1. For all fitting requiring Loctite 565 Thread Sealant (636-666-00), apply Loctite 565 Thread Sealant to threads and spread evenly to cover a minimum of five full threads, but skip the first thread .
   Wear gloves when handling the Loctite to avoid skin irritation

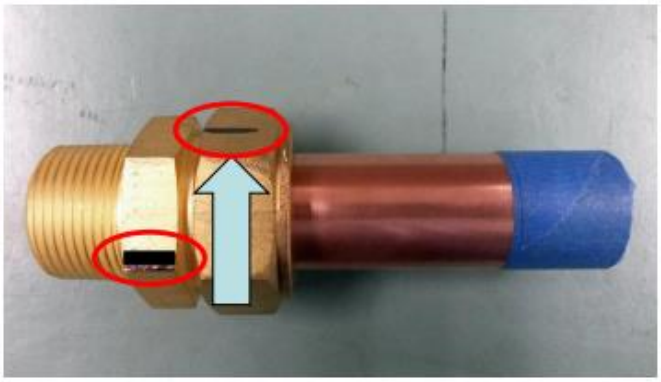
MAN: CERTIFIED OPERATOR, TECHNICIAN

NOTES (continued)

2. Process for tightening of Pipe thread and compression thread fittings
   A. Hand tighten the parts. In the case of a compression fit make sure the pipe is fully inserted into the fitting
   B. Using a “Sharpie” style black pen, make a mark in line on both components

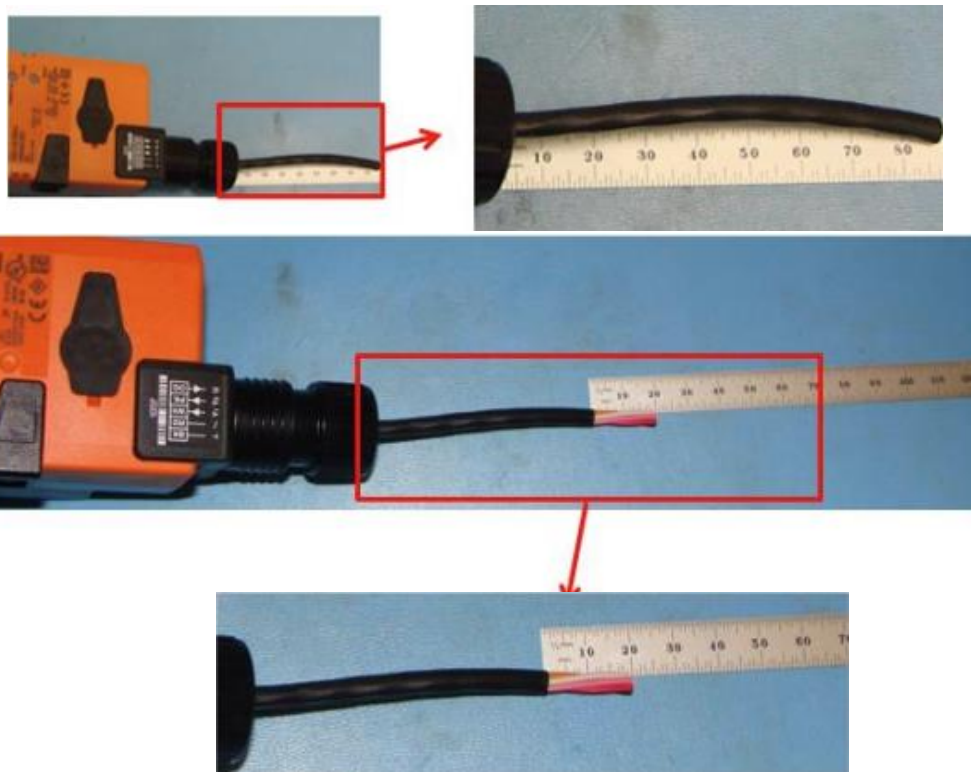


   C. Tighten the fitting to the specification stated in the specific assembly instruction example: “Tighten 1 and ¼ turns”



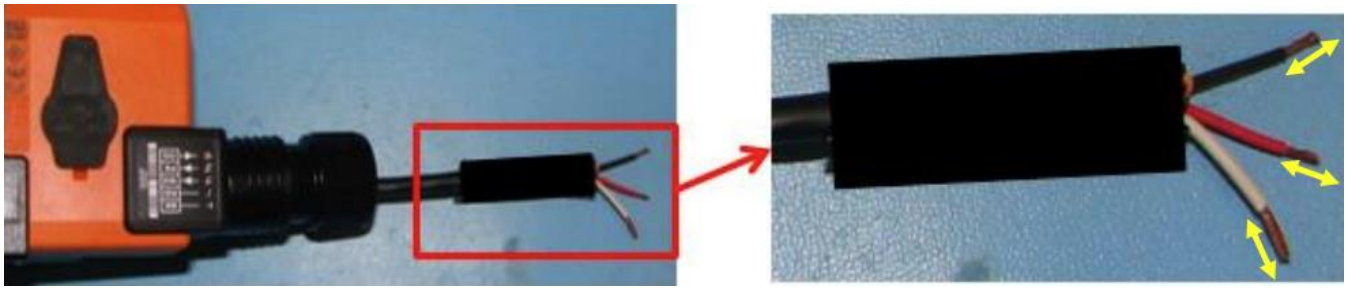
MAN: CERTIFIED OPERATOR, TECHNICIAN

Obtain Belimo Valve 606-259-12
Cut Cable to an overall length of 600mm
Strip the cable insulation jacket 40mm, exposing the individual wires
There should be 4 wires with the corresponding numbers and colors: (1) black, (2) red, (3) black, (5) orange



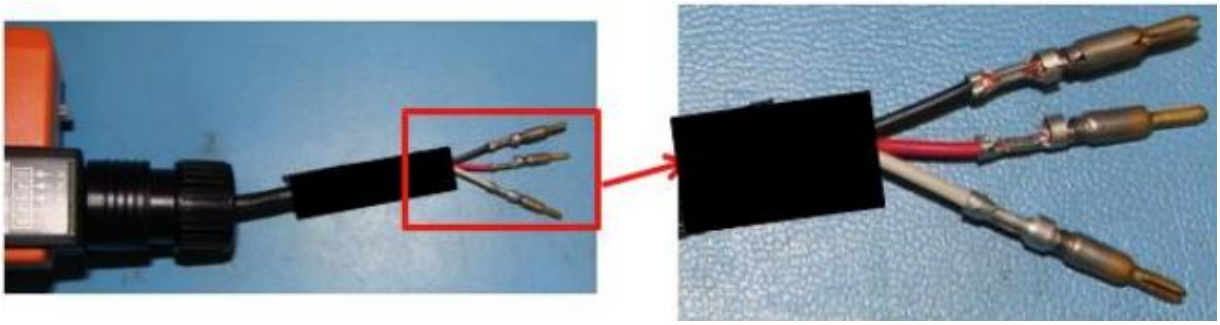
MAN: CERTIFIED OPERATOR, TECHNICIAN

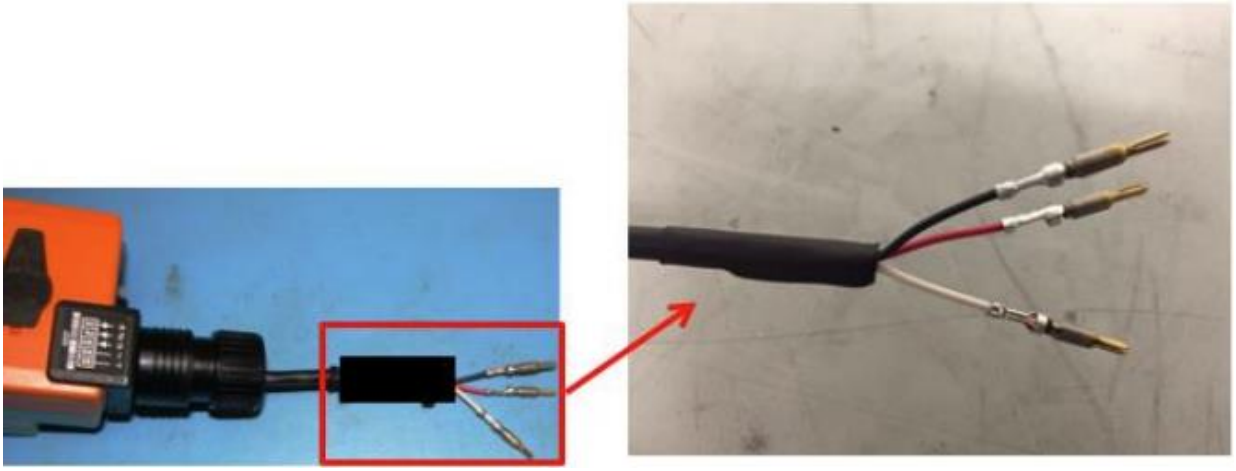
Bend the (5) orange wire back
Obtain heat shrink 660-050-00 and place it over the cable as shown, covering the orange cable just past the bend in the wire
Shrink heat shrink into place
Strip the remaining 3 individual wires 5mm



MAN: CERTIFIED OPERATOR, TECHNICIAN

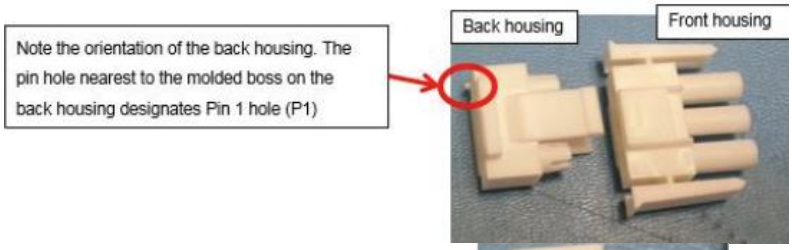
Crimp pin connectors 634-075-01 on to the x3 wires using a standard crimping tool

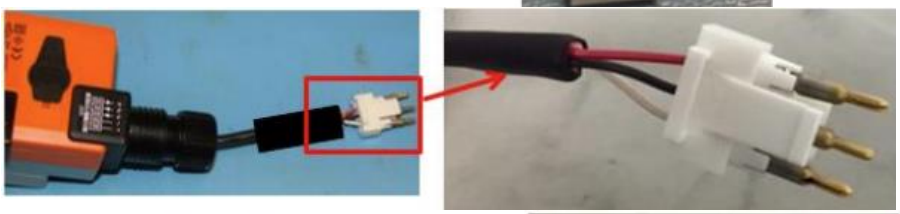


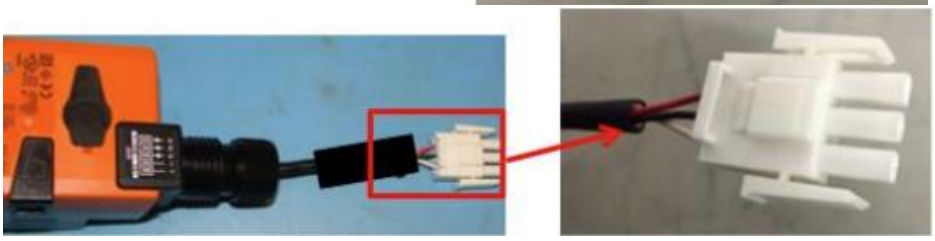


MAN: CERTIFIED OPERATOR, TECHNICIAN

Obtain the 3 Pin Plug 359-494-03 (includes front and back housings)
Take the back housing of the receptacle and insert the pin connectors into the holes until they are fully seated.
Install the front housing onto the back housing by snapping into place

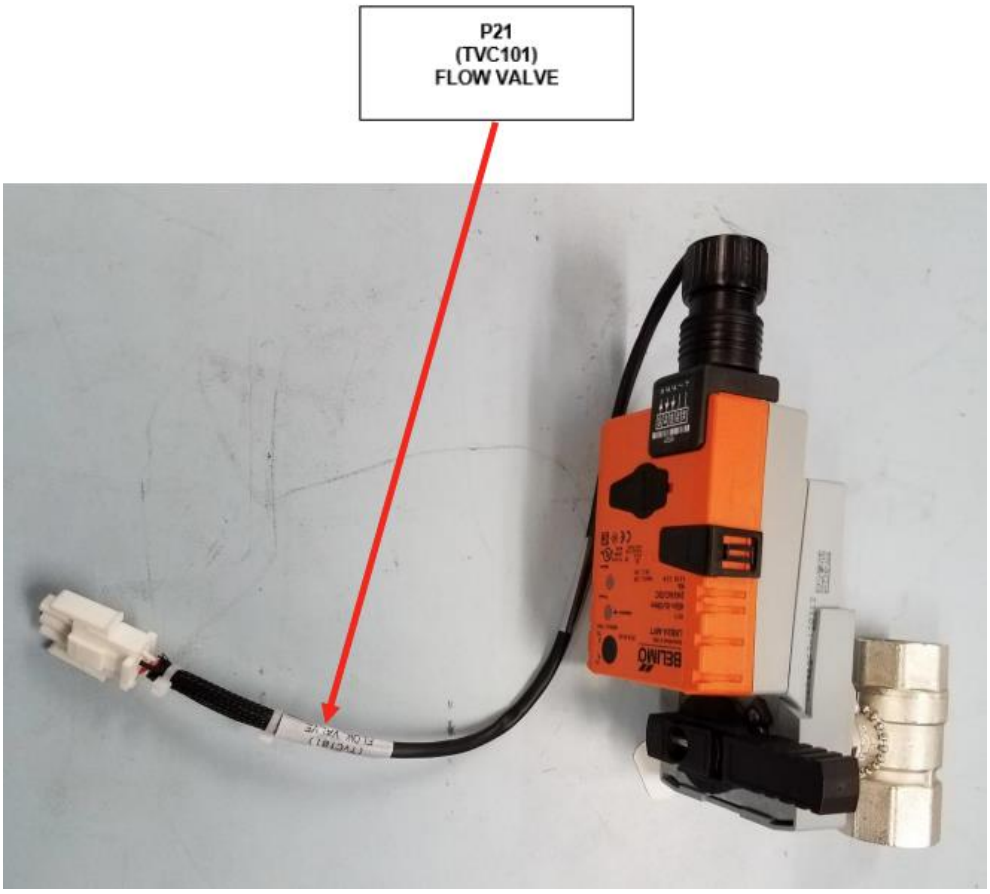






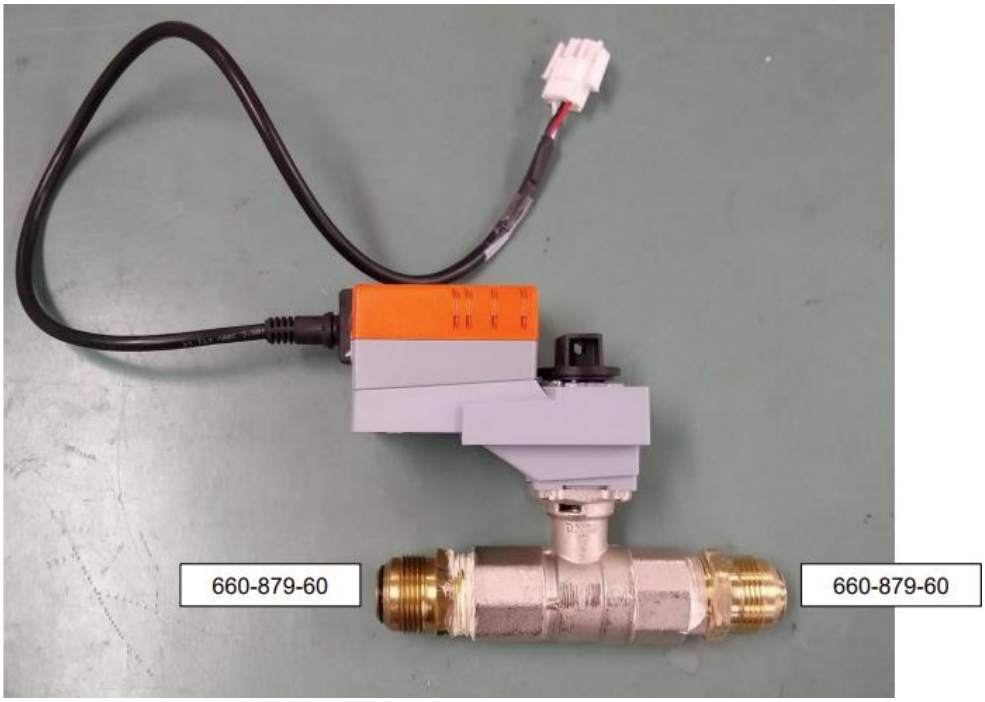
MAN: CERTIFIED OPERATOR, TECHNICIAN

Create the label below and place it in the location shown



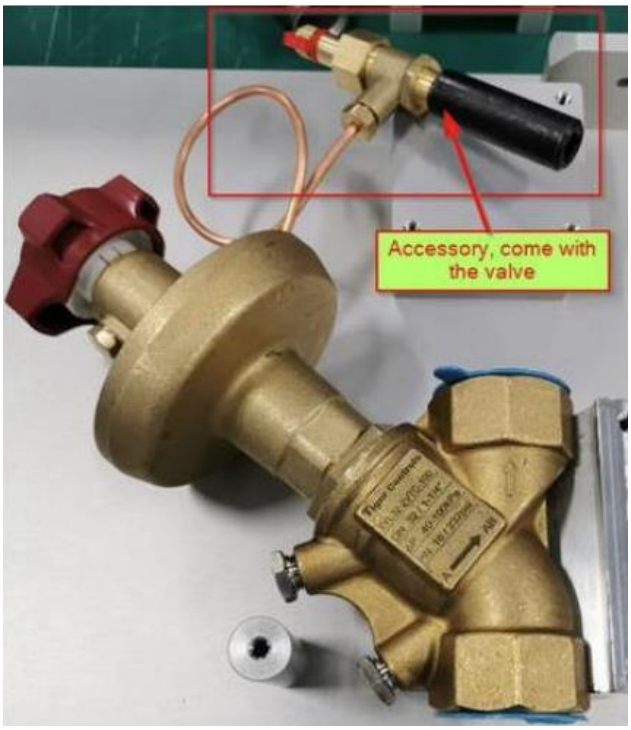
MAN: CERTIFIED OPERATOR, TECHNICIAN

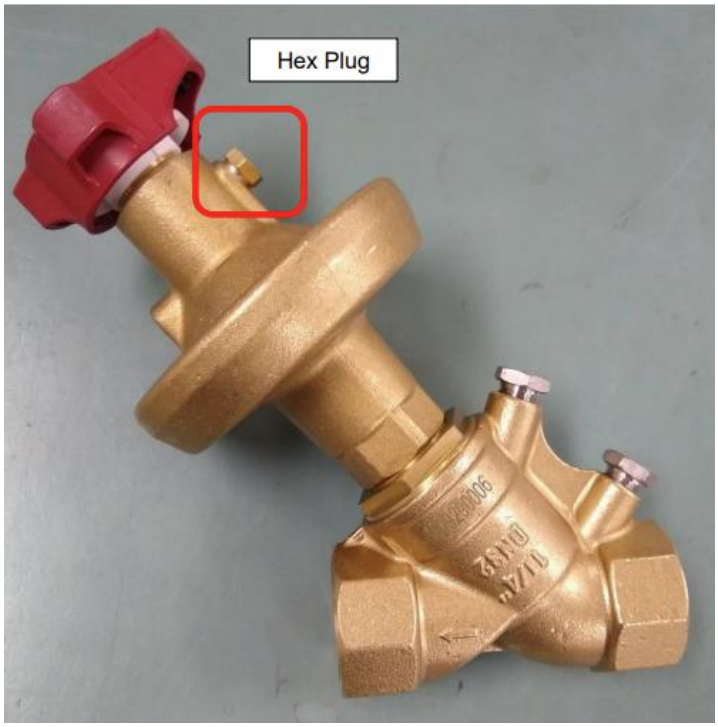
Inspect threads on fitting 660-879-60 x2 for any damage
Place Belimo valve in vice grip
Apply thread sealant Loctite 565 to R thread side of fitting 660-879-60
Use Gloves, follow basic practices mentioned earlier in MMI
Hand thread fitting to hand tight to Valve 606-259-12
Mark positions on fitting and valve
Using a 156 Nm torque tool with a 1 3/8” wrench, Tighten fitting, ensuring the fitting rotates 2-3 turns when it reaches torque
Repeat other side



MAN: CERTIFIED OPERATOR, TECHNICIAN

Acquire Tiger Valve 662-520-00
Remove and discard extra accessory that comes with the valve
Torque hex plug on Tiger Valve 662-520-00
12mm Hex
Torque to 3.3 Nm
Note: this plug should already be tight and this torque step is just for verification



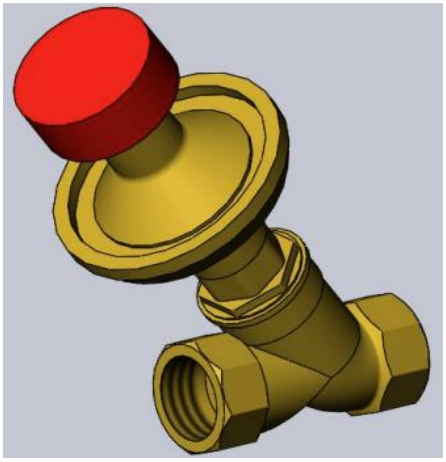


MAN: CERTIFIED OPERATOR, TECHNICIAN

Apply Molykote 111 to x2 O-rings 470-355-08
Place O-ring over fitting 660-879-70
Place brass ring 660-680-03 over O-ring
Apply Molykote 111 to threads on fittings
Place Tiger Valve 662-520-00 in to vice grip
Don't Clamp on brass ring!
Hand thread fitting to Tiger Valve 662-520-00
Use 50mm Crowsfoot wrench and torque to 156 Nm
Warning: this is a high torque!
Repeat other side of Valve

660-680-03 Brass ring
660-879-70 Fitting
470-355-08 O-ring

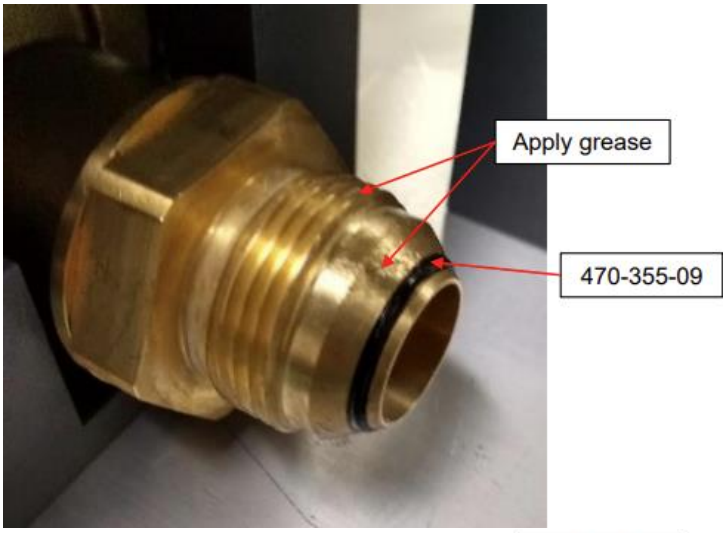
662-520-00

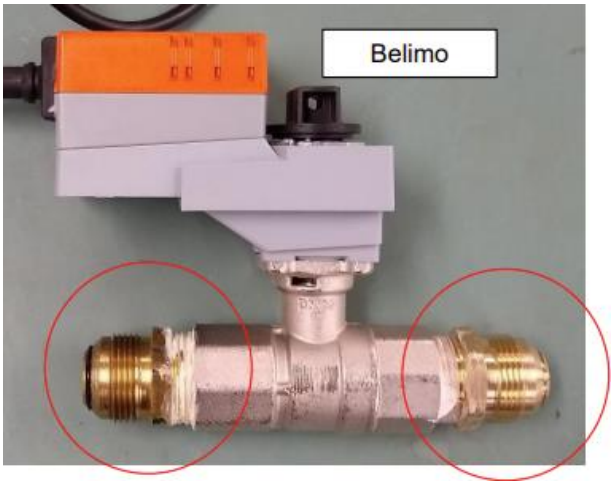


660-680-03 Brass ring
660-879-70 Fitting
470-355-08 O-ring

MAN: CERTIFIED OPERATOR, TECHNICIAN

Apply Molykote 111 to O-ring 470-355-09 x5
Install O-ring to fittings 660-879-50 on flow meter and valve, and lone fitting 660-879-80
Apply Molykote 111 to fitting flare and a couple threads



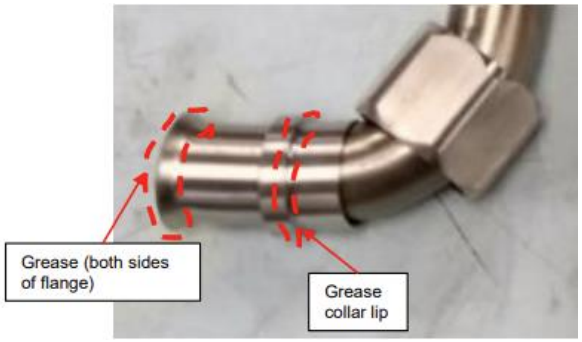


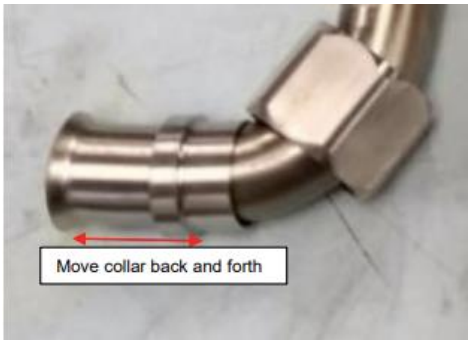
Safety Tips: N/A

TQC: N/A

MAN: CERTIFIED OPERATOR, TECHNICIAN

Apply Molykote 111 to the flange and collar lip on pipes 658-802-02, 658-802-03, and 658-802-06, both ends of pipe.
Move Collars up and down to ensure they are floating and not stuck

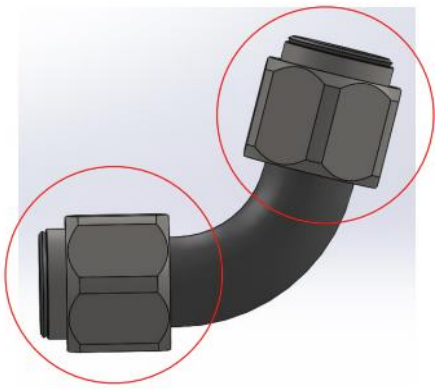




658-802-02

658-802-03

658-802-06

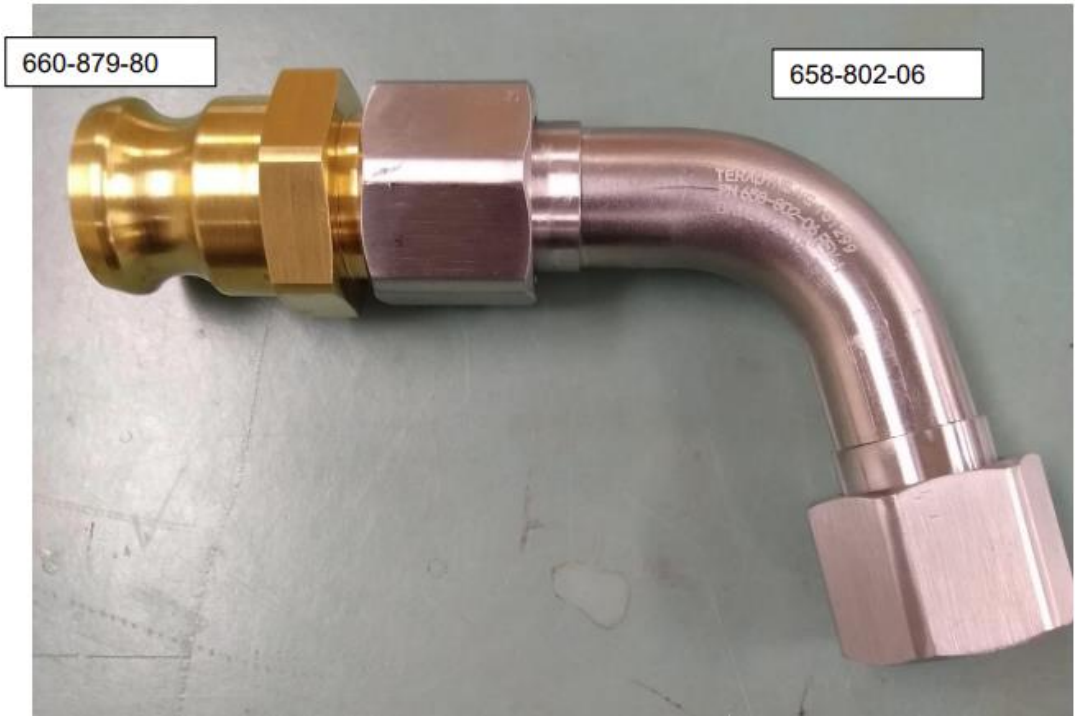


MAN: CERTIFIED OPERATOR, TECHNICIAN

Check that pipe 658-802-06 and fitting 660-879-80 have Molykote 111 applied
Place fitting 660-879-80 into vice grip
Thread fitting and pipe 658-802-06 together to hand tight
Use 1-1/2" Crowsfoot wrench and torque to 156 Nm

Warning: this is a high torque!

Hold Pipe steady while tightening

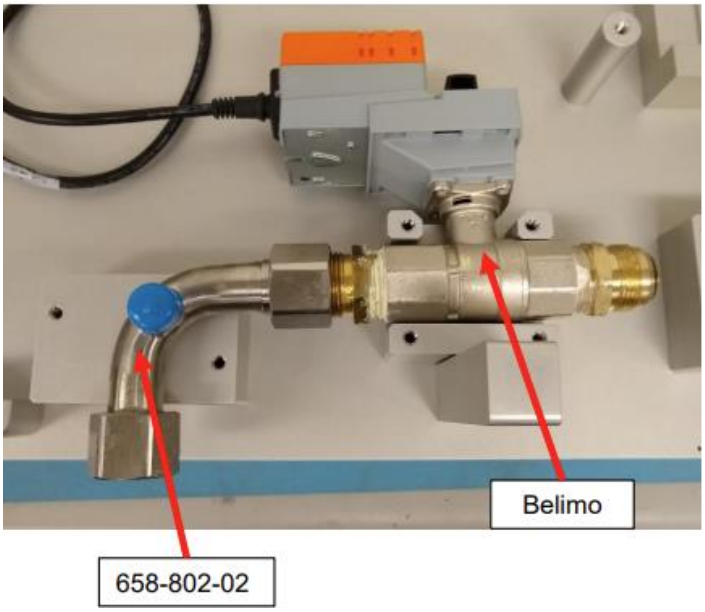


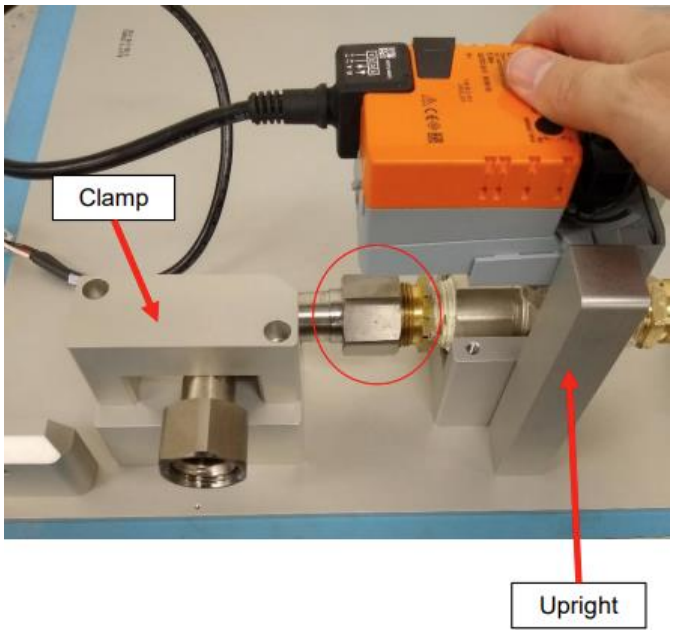
MAN: CERTIFIED OPERATOR, TECHNICIAN

Place Belimo sub-assembly in Orientation fixture (Note position of feature on pipe)
Place pipe 658-802-02 in orientation fixture
Thread fitting and pipe together to hand tight, then back off 1/4 rotation.
Rotate Belimo vertical and rest it against the upright
Hand tight pipe.
Clamp pipe to Orientation Fixture. Torque screws to 1.6 Nm.
Use 1-1/2" Crowsfoot wrench and torque to 156 Nm

Warning: this is a high torque!

Use 1 3/8" wrench on the Belimo's fitting to secure it in place



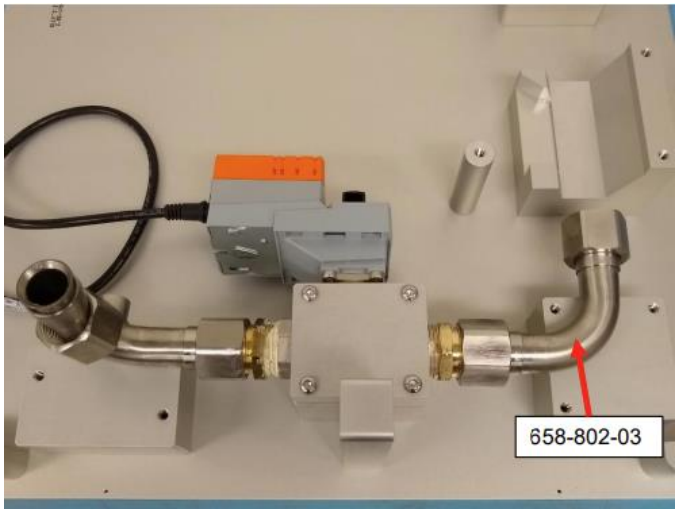


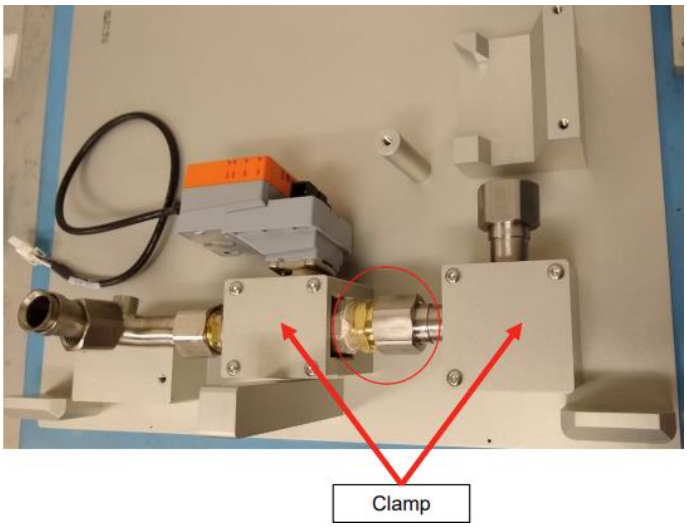
MAN: CERTIFIED OPERATOR, TECHNICIAN

Remove Orientation Fixture Clamp from Pipe
Rotate sub-assembly by resting Belimo valve on Orientation Fixture base
Make sure grease is applied to unused fitting and O-ring on Belimo Valve
Thread fitting and pipe 658-802-03 together to hand tight, then back off 1/4 rotation
Adjust pipe so that it is resting on orientation fixture block.
Hand tight pipe to Belimo Valve
Clamp Belimo and pipe to orientation fixture. Torque screws to 1.6 Nm
Use 1-1/2" Crowsfoot wrench and torque to 156 Nm

Warning: this is a high torque!

Use 1 3/8" wrench on Belimo valve fitting to hold it in place

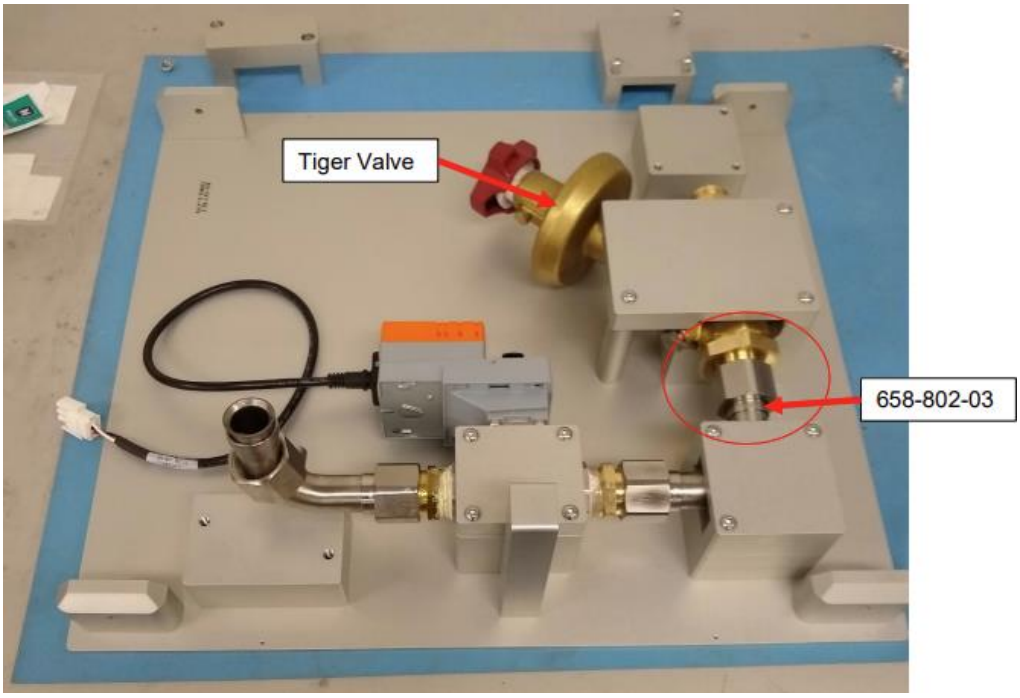




MAN: CERTIFIED OPERATOR, TECHNICIAN

Make sure grease is applied to unused fitting and O-ring on the Tiger Valve
Thread pipe 658-802-03 to Tiger Valve fitting to hand tight, then back off 1/4 rotation
Adjust pipe and Tiger valve so they match orientation as shown (may need to undo clamps on Orientation Fixture).
Hand tight pipe to Tiger valve
Clamp Tiger Valve and pipe to orientation fixture. Torque screws to 1.6 Nm
Use 1-1/2" Crowsfoot wrench and torque to 156 Nm

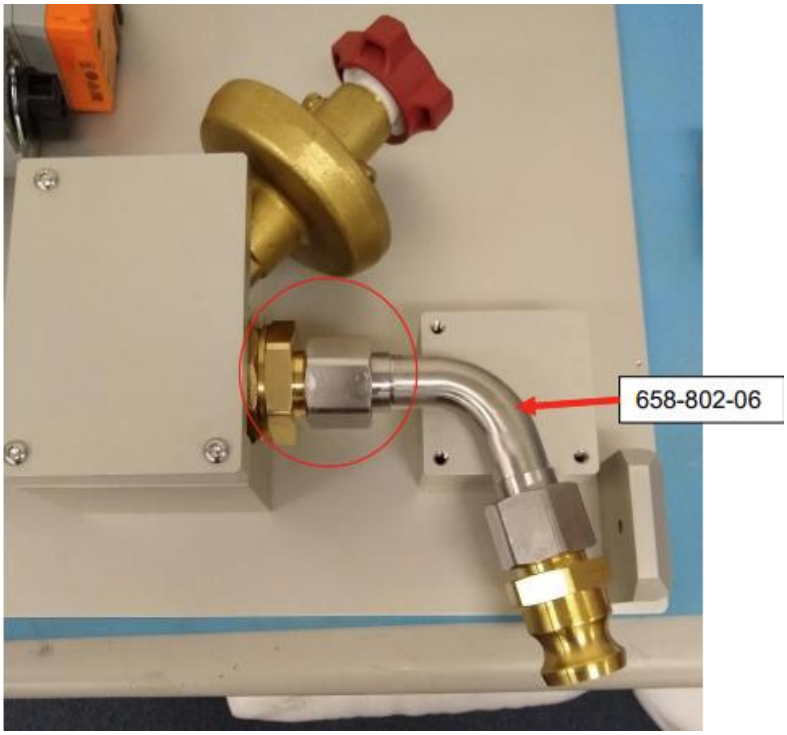
Warning: this is a high torque!



MAN: CERTIFIED OPERATOR, TECHNICIAN

Make sure grease is applied to unused fitting and O-ring on Tiger Valve
Thread fitting and pipe 658-802-06 together to hand tight, then back off 1/4 rotation.
Adjust pipe so it is resting flat on Orientation Fixture
Hand tight pipe to Tiger Valve, keeping pipe flat on Orientation Fixture.
Clamp pipe to Orientation Fixture. Torque screws to 1.6 Nm.
Use 1-1/2" Crowsfoot wrench and torque to 156 Nm

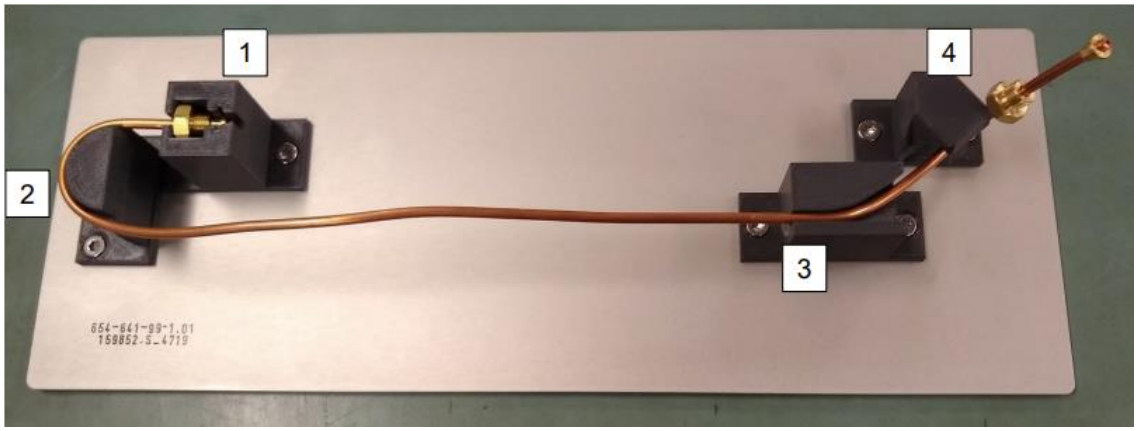
Warning: this is a high torque!



MAN: CERTIFIED OPERATOR, TECHNICIAN

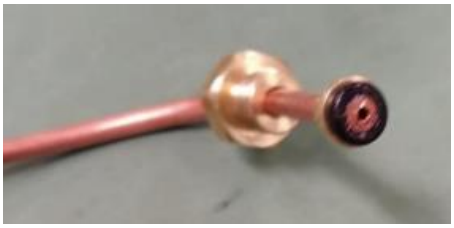
Use Sense Line Orientation Bend Fixture 654-641-99 on Sense line / capillary tube 667-283-00

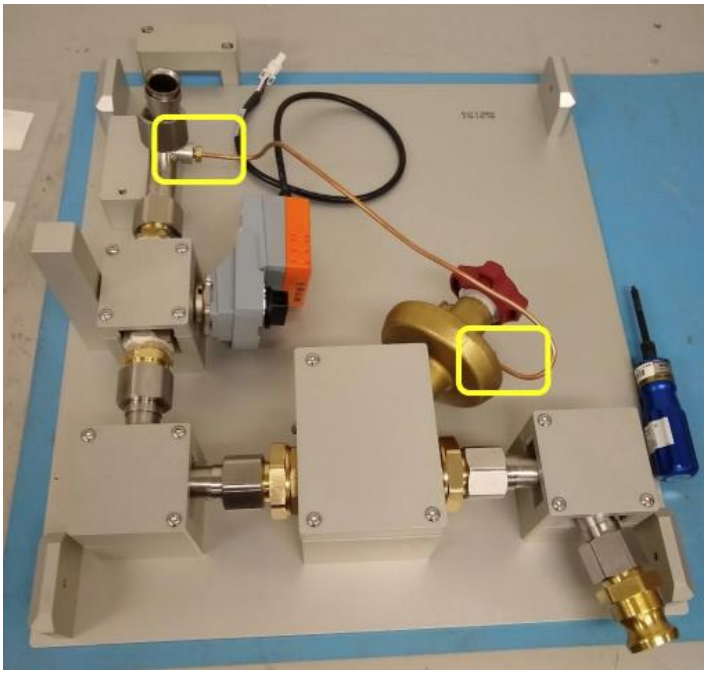
1. Place one end of sense line in start location
2. Bend line around turn
3. Bend line around second turn. Hold line steady
4. Bring line underneath last bend and follow up curve trajectory

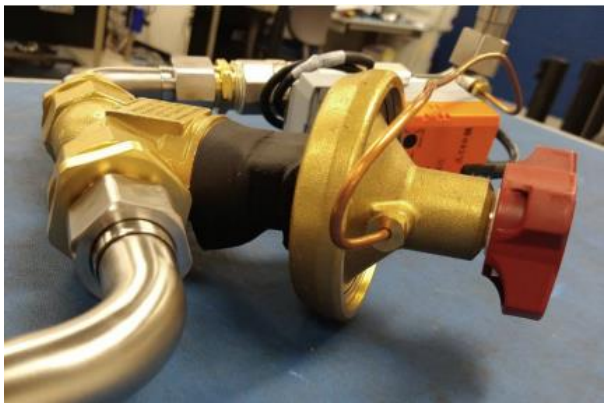


MAN: CERTIFIED OPERATOR, TECHNICIAN

Apply Molykote 111 Grease to O-rings 666-544-00 x2
Apply Molykote 111 Grease to threads on Sense Line
Place x1 O-ring on one end of Sense Line
Install sense line to Water Return subassembly, start with hand thread
    14mm Hex
    Torque to 5.6 Nm
Place other O-ring on other end of Sense Line
    It is ok to slightly bend the line to align parts
Install other end of sense line, start with hand thread
    14mm Hex
    Torque to 5.6 Nm







ASSEMBLY COMPLETE
MOVE ON TO LEAK TEST